Clinical tools for cholera treatment facilities

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Sources and intended use

This collection of tools contains re-presentation of existing WHO guidance for cholera, which are available at: https://www.who.int/emergencies/outbreak-toolkit/disease-outbreak-toolboxes/cholera-outbreak-toolbox.

The tools here are intended to enable easy access to clinical resources during cholera outbreaks. They focus on patient management in cholera treatment facilities, and complement other WHO resources for out-patient management of cholera and diarrhoeal disease.

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With thanks to the Global Task Force on Cholera Control case management working group.

Cholera patient treatment flowchart

Case definition suspected cholera: *Outside of cholera outbreak:* any patient ≥2 years presenting with acute watery diarrhoea and severe dehydration or dying from acute watery diarrhoea. *Where a cholera outbreak is declared:* any person presenting with or dying from acute watery diarrhoea.

START: Does the patient fit the case definition?

No → Manage outside of cholera pathway

Yes → **Assess for shock / severe dehydration**

Patient has one or more danger sign(s):
- Lethargic or unconscious
- Absent or weak pulse
- Respiratory distress

OR patient has at least 2 of the following:
- Not able to drink or drinks poorly
- Skin pinch goes back very slowly
- Sunken eyes

Severe dehydration or shock → **PLAN C**

Give intravenous **Ringer's lactate**…

Age	Ringer's lactate (mL)
under 1 year	30 mL / kg in 60 minutes, then 70 mL / kg over the next 5 hours
1 year +	30 mL / kg in 30 minutes, then 70 mL / kg over the next 2½ hours

Be alert for hypoglycaemia, and treat rapidly with intravenous dextrose

Reassess every 15–30 minutes
- Awake and alert ***AND***
- Able to drink ***AND***
- Improved pulse strength

Yes → *Some dehydration* → **PLAN B**

No → **Assess for some dehydration**

Patient has at least 2 of the following:
- Irritable or restless
- Sunken eyes
- Rapid pulse
- Thirsty (drinks eagerly)
- Skin pinch goes back slowly

Give oral rehydration solution (**ORS**) **over 4 hours, 75 mL/kg:**

Age	Weight (kg)	ORS (mL)
< 4 months	*under 5 kg*	200–400
4–11 months	*5–7.9 kg*	400–600
12–23 months	*8–10.9 kg*	600–800
2–4 years	*11–15.9 kg*	800–1200
5–14 years	*16–29.9 kg*	1200–2200
15 years +	*30 kg +*	2200–4000

Reassess at least every hour

Severe dehydration ***OR*** Severe vomiting → Yes

Awake and alert ***AND*** Able to drink ***AND*** Improved pulse strength → Yes → **PLAN A**

No → *No dehydration*

No hydration found
- Awake and alert
- Normal pulse
- Normal thirst
- Eyes not sunken
- Skin pinch normal

PLAN A

Give oral rehydration solution (**ORS**) **over 4 hours** according to age:

Age	ORS (mL)
< 24 months	50–100
2–9 years	100–200
10 years +	As much as wanted

Reassess and consider discharge if
- Has no signs of dehydration ***AND***
- Takes ORS without vomiting ***AND***
- Has no watery stools for 4 hours ***AND***
- Is able to walk without assistance ***AND***
- Is passing urine

Reference (scan QR code)

1) Global Task Force for Cholera Control (GTFCC)

Antibiotics for all patients with severe dehydration, pregnant women and the elderly:
- *Preferred is doxycycline single dose (200 mg for under 12 years, or 300 mg for older children and adults.*
- *Alternative is azithromycin single dose (20 mg / kg for under 12 years, or 1 g for older children and adults.*

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Evaluating dehydration and admission criteria

	Signs	Treatment	Admit
Severe dehydration	**One or more danger signs:** • Lethargic or unconscious • Absent or weak pulse • Respiratory distress **OR at least 2 of the following:** • Sunken eyes • Not able to drink or drinks poorly • Skin pinch goes back very slowly	Plan C	Yes
Some dehydration	**No danger signs AND at least 2 of the following:** • Irritable or restless • Sunken eyes • Rapid pulse • Thirsty (drinks eagerly) • Skin pinch goes back slowly	Plan B	Yes
No dehydration	• Awake and alert • Normal pulse • Normal thirst • Eyes not sunken • Skin pinch normal	Plan A	No

Treatment for **Severe dehydration**

Start intravenous fluids immediately

Plan C

Age	first give **30 mL / kg in**	then give **70 mL / kg in**
Infants *(< 12 months)*	1 hour	5 hours
All others *(12 months +)*	30 minutes	2½ hours

Start intravenous fluid immediately
Ringers lactate is the first choice

Use large cannulas
or put 2 IV lines if large volumes are to be administered rapidly

Write the start time on the IV bag
to help track how quickly fluids are being given

Reassess
at least every 15 to 30 minutes

Watch for signs of hypoglycaemia
and treat rapidly

Give ORS to replace ongoing losses
as soon as the patient can drink safely

Zinc - give for 10 days
In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

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Treatment to
Maintain hydration

Plan C (ORS)

Age	ORS after each loose stool
< 24 months	50–100 mL
2–9 years	100–200 mL
10 years +	As much as wanted

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Treatment for
Some dehydration

Plan B (ORS)

1 Give ORS
over 4 hours

2 Reassess
at least every hour

3 Give additional ORS
to replace ongoing losses in addition to fluids in the table

Age	Weight (kg)	ORS in 4 hours
< 4 months	< 5 kg	200–400 mL
4 – 11 months	5 – 8 kg	400–600 mL
12 – 23 months	8 – 12 kg	600–800 mL
2 – 4 years	12 – 16 kg	800–1200 mL
4 – 14 years	16 – 30 kg	1200–2200 mL
15 + years	30+ kg	2200–4000 mL

Zinc - give for 10 days
In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

Antibiotics
All pregnant women and elderly patients should receive antibiotics regardless of their hydration status or treatment plan

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Treatment to
Maintain hydration

Plan B

Age	ORS after each loose stool
< 24 months	50–100 mL
2–9 years	100–200 mL
10 years +	As much as wanted

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Treatment for
No sign of dehydration:

Plan A (ORS)

Age	ORS after each loose stool	Give enough ORS packets for
< 24 months	50 – 100 mL	500 mL / day
2 – 9 years	100 – 200 mL	1 L / day
10 years +	As much as wanted	2 L / day

Zinc - give for 10 days

In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

Antibiotics

All pregnant women and elderly patients should receive antibiotics regardless of their hydration status or treatment plan

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Cholera fluids quick reference

Children >1 year old, and adults

(plan C is 100mL/kg in **3 hours**)

Weight (kg)	Total volume	30 mL/kg* bolus Ringers over 30 mins		70 mL/kg Ringers over 2½ hours
10–13 kg	1000 mL	250 mL in 30 mins	(≈ 8 mL/min)	750 mL in 2½ hours
13–15 kg	1200 mL	300 mL in 30 mins	(≈ 10 mL/min)	900 mL in 2½ hours
15–17 kg	1500 mL	300 mL in 30 mins	(≈ 10 mL/min)	1200 mL in 2½ hours
17–19 kg	1700 mL	400 mL in 30 mins	(≈ 15 mL/min)	1300 mL in 2½ hours
19–20 kg	2000 mL	600 mL in 30 mins	(≈ 20 mL/min)	1400 mL in 2½ hours
21–23 kg	2200 mL	700 mL in 30 mins	(≈ 25 mL/min)	1500 mL in 2½ hours
24–26 kg	2500 mL	800 mL in 30 mins	(≈ 25 mL/min)	1700 mL in 2½ hours
27–29 kg	2800 mL	800 mL in 30 mins	(≈ 25 mL/min)	2000 mL in 2½ hours
30–31 kg	3000 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2000 mL in 2½ hours
32–34 kg	3300 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2300 mL in 2½ hours
35–38 kg	3500 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2500 mL in 2½ hours
39–43 kg	4000 mL	1200 mL in 30 mins	(≈ 40 mL/min)	2800 mL in 2½ hours
44–49 kg	4500 mL	1500 mL in 30 mins	(≈ 50 mL/min)	3000 mL in 2½ hours
50–54 kg	5000 mL	1500 mL in 30 mins	(≈ 50 mL/min)	3500 mL in 2½ hours
≥ 55 kg	6000 mL	2000 mL in 30 mins	(≈ 70 mL/min)	4000 mL in 2½ hours

Infants (under 1 year, plan C 100mL/kg in **6 hours**)

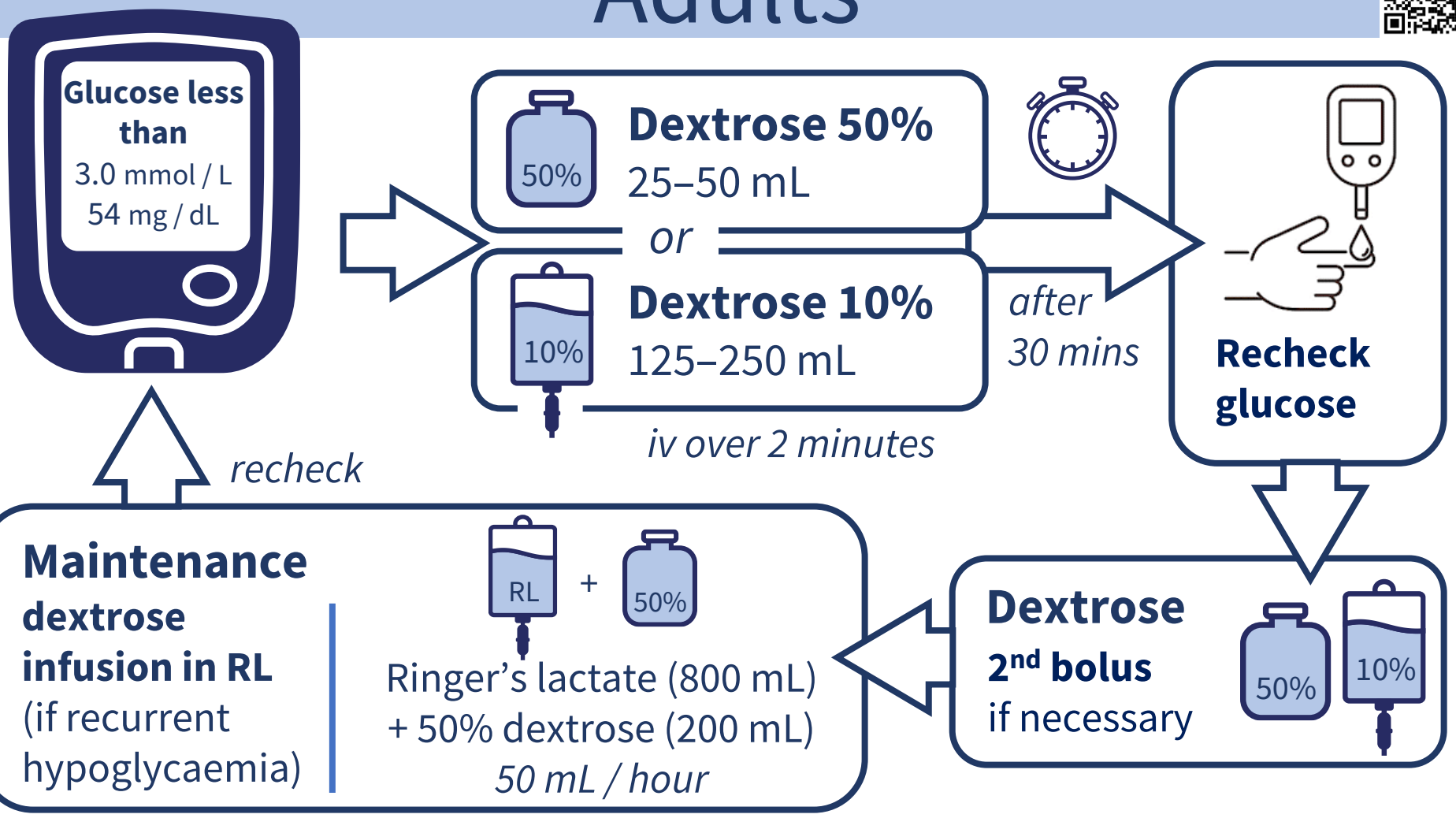
Age (years)	Weight (kg)	Total volume	30 mL/kg* Ringers over 1 hour	70 mL/kg* Ringers over 5 hours
<1 month	3–4 kg	350 mL	105 mL in 1 hour	245 mL in 5 hours
1–2 months	4–5 kg	450 mL	135 mL in 1 hour	315 mL in 5 hours
2–3 months	5–6 kg	550 mL	165 mL in 1 hour	385 mL in 5 hours
3–4 months	6–7 kg	650 mL	195 mL in 1 hour	455 mL in 5 hours
4–7 months	7–8 kg	750 mL	225 mL in 1 hour	525 mL in 5 hours
7–10 months	8–9 kg	850 mL	255 mL in 1 hour	595 mL in 5 hours
10–12 months	9–10 kg	950 mL	285 mL in 1 hour	665 mL in 5 hours

**Fluid should be administered through an appropriate infusion set (1 mL = 60 drops; 150 mL chamber for paediatrics)*

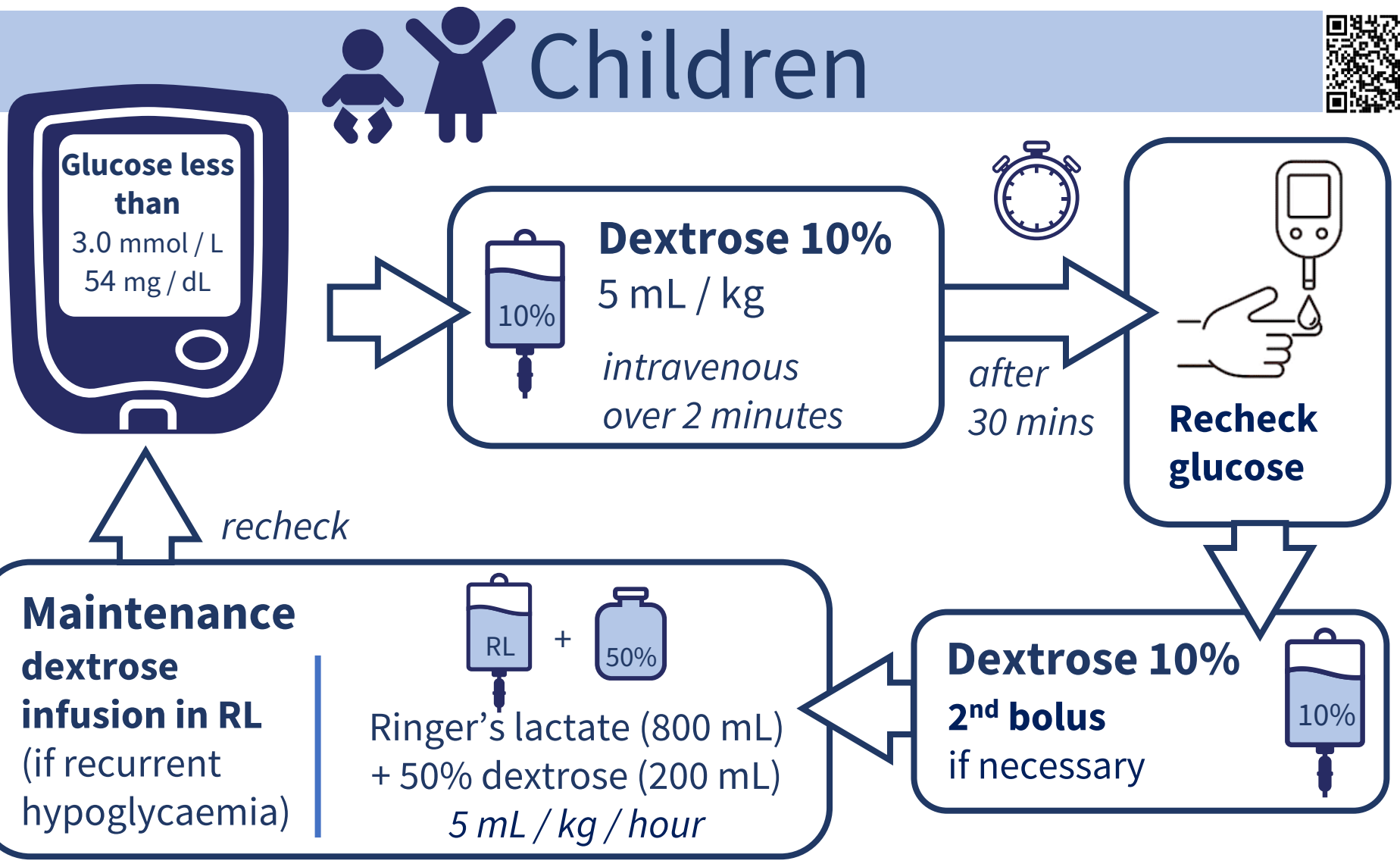
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Hypoglycaemia quick reference

Adults



Children



If no intravenous access, 2-5ml 50% dextrose fluid can be placed directly the cheek using a syringe. Do not use a needle or spoon.

References (scan QR code)

1) *WHO-ICRC Basic Emergency Care*
2) *Pocket book of hospital care for children*

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Discharge assessment

- ❑ Ensure patients and their caregivers understand that anyone with at least 3 episodes of diarrhoea in 24 hours should seek treatment.
- ❑ Ensure all patients and their caregivers understand how to prevent cholera before discharge.
- ❑ Provide at least 4 packets of ORS for all patients who are discharged to use at home.

  The packets can also be used for any family members or neighbours how begin to have diarrhoea.
- ❑ For paediatric patients give sufficient zinc to complete 10 days treatment.

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https://www.who.int